

Creature v06 Hardware Build and Bring-Up Checklist

Date: _____ Version: _____

Test Sketch Version: _____ *version of the bring-up test sketch, not the v06 firmware*

Reference: Basic Breadboard Placement

Confirmed pins, taken from current firmware and wiring docs. Where a pin is not yet assigned, it is marked TBD. Fill it in once chosen, do not guess on the bench.

ESP32-S3-DevKitC-1-N8R8

USB-C power and data.
3V3 and GND rails feed every sensor below.
Onboard status NeoPixel: internal GPIO38, no external wiring.
Octal PSRAM (N8R8) reserves some GPIOs for memory. Check the DevKitC-1 pinout before assigning new pins. Avoid GPIO0, 3, 45, 46 (strapping).

I2C bus, shared: SDA GPIO8, SCL GPIO9

BH1750 (light)

VCC -> 3V3, GND -> GND.
ADDR -> GND, address 0x23.

MPU-6050 / ICM20689 (motion)

VCC -> 3V3, GND -> GND.
Same I2C bus, address 0x68.
Use the finani ICM20689 driver, not a generic MPU6050 library.

BME280 (weather)

VCC -> 3V3, GND -> GND.
Same I2C bus, address 0x76 (or 0x77).

SK6812 RGBW strip (~16 px)

5V and GND from the board, common ground.
DIN -> GPIO ____ (not yet assigned in firmware).
330-470 ohm resistor inline on DIN.
1000 uF capacitor across 5V/GND at the strip.

I2S mic: SCK GPIO5, WS GPIO6, SD GPIO7

INMP441 (sound)

VDD -> 3V3, GND -> GND.
L/R -> GND (left channel select).

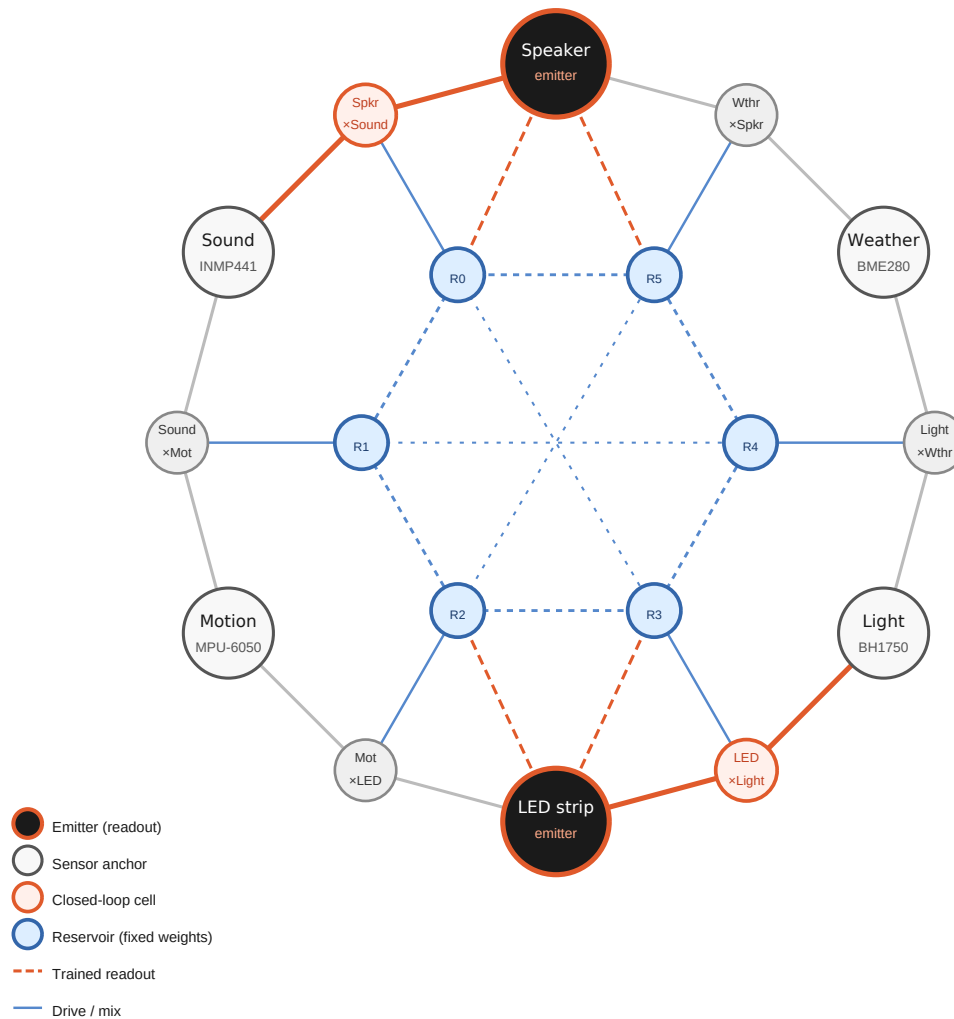
MAX98357A + speaker (voice)

3.3-5V and GND from the board.
BCLK -> GPIO ____, LRC -> GPIO ____, DIN -> GPIO ____ (not yet assigned in firmware).

Notes:

Reference: v06 Field Ring Layout

This is the field's conceptual ring and reservoir topology, sensor and emitter anchors, in-between cells, and the inner reservoir. It is not a wiring diagram. For physical pins, use the breadboard placement page above.



Phase 0: Build the Physical Body

Sensor Board

- ☐ ESP32-S3 mounted
- ☐ BH1750 mounted in intended location
- ☐ INMP441 mounted in intended location
- ☐ MPU-6050 (ICM20689-compatible) mounted in intended location
- ☐ BME280 mounted in intended location
- ☐ Power rails established
- ☐ Ground rails established
- ☐ Wires routed neatly
- ☐ Connections labeled

Notes:

Expression Board

- ☐ MAX98357A amplifier mounted
- ☐ Speaker mounted
- ☐ SK6812 strip mounted
- ☐ SK6812 data-line resistor installed (330-470 ohm)
- ☐ SK6812 1000 uF capacitor installed across 5V/GND
- ☐ Common ground connected
- ☐ Expression board can be repositioned

Notes:

Phase 1: Electrical Verification

- ☐ ESP32-S3 powers on
- ☐ No unexpected heat
- ☐ No shorts between VCC and GND
- ☐ All power connections verified with multimeter
- ☐ All ground connections verified
- ☐ INMP441 L/R pin confirmed tied to GND (left channel select)
- ☐ Wiring visually inspected

Notes:

Phase 2: I2C Bus Scan

☐ I2C scanner uploaded successfully

BH1750 found at: _____ *ADDR tied to GND, expect 0x23*

MPU-6050 found at: _____ *expect 0x68*

BME280 found at: _____ *expect 0x76 or 0x77*

INMP441 will not appear here. It is I2S, not I2C. Expected, not a fault.

Notes:

Phase 3: Individual Hardware Tests

BH1750 Light Sensor

Test sketch used: _____

- ☐ Lux changes when covered
- ☐ Lux increases under flashlight or lamp
- ☐ Lux approaches zero in darkness
- ☐ Readings stable

Notes:

INMP441 Microphone

Test sketch used: _____

- ☐ RMS values present
- ☐ RMS increases when speaking
- ☐ RMS increases when clapping
- ☐ RMS decreases in quiet room
- ☐ No clipping
- ☐ No constant high readings

Notes:

MPU-6050 Motion Sensor

Test sketch used: _____

- ☐ Confirmed using ICM20689-compatible driver (finani/ICM20689), not a generic MPU6050 library
- ☐ Values change when moved
- ☐ Values settle when stationary
- ☐ Motion scalar increases when tapped
- ☐ Motion scalar decreases when still
- ☐ No obvious drift

Notes:

BME280 Weather Sensor

Test sketch used: _____

- ☐ Temperature plausible
- ☐ Pressure plausible
- ☐ Humidity plausible
- ☐ Readings stable
- ☐ Temperature responds to warm hand
- ☐ Pressure remains nonzero

Notes:

SK6812 RGBW LED Strip

Test sketch used: _____

- ☐ First pixel responds
- ☐ All pixels illuminate
- ☐ Red works
- ☐ Green works
- ☐ Blue works
- ☐ White works
- ☐ No dead pixels
- ☐ No flickering
- ☐ Brightness controllable

Notes:

MAX98357A Speaker

Test sketch used: _____

- ☐ Test tone plays
- ☐ Low volume works
- ☐ Medium volume works
- ☐ High volume works
- ☐ No distortion
- ☐ No constant hiss

Notes:

Phase 4: Loop Placement Validation

This phase validates physical placement and raw sensing only. Closing the firmware loop happens later, in firmware integration, loose and tightened slowly.

Light Loop

- ☐ BH1750 detects strip brightness
- ☐ Sensor does not saturate
- ☐ No runaway between strip and sensor (light driving sensor driving light)
- ☐ Distance tested
- ☐ Angle tested
- ☐ Final position marked

Distance (cm): _____ Angle (degrees): _____

Notes:

Sound Loop

- ☐ INMP441 detects speaker output
- ☐ No immediate feedback runaway
- ☐ Distance tested
- ☐ Angle tested
- ☐ Final position marked

Distance (cm): _____ Angle (degrees): _____

Notes:

Phase 5: Stability Test (30-60 Minutes)

- ☐ No sensor disconnects
- ☐ No I2C lockups
- ☐ No WiFi crashes
- ☐ No unexpected reboots
- ☐ No overheating
- ☐ Sensor readings remain stable

Notes:

Ready for Firmware Integration

- ☐ Physical body assembled
- ☐ Power verified
- ☐ I2C verified
- ☐ BH1750 complete
- ☐ INMP441 complete
- ☐ MPU-6050 complete
- ☐ BME280 complete
- ☐ SK6812 complete
- ☐ MAX98357A complete
- ☐ Light loop validated
- ☐ Sound loop validated
- ☐ Stability test passed

Final Sign-Off

- ☐ Creature body hardware is ready for v06 firmware integration.

Git tag created: _____ *e.g. v06-hw-bringup*

Date: _____ Signature: _____